Mahesh Chitakoti

Bengaluru, India | +91 94817 77227 | maheshchitkoti@gmail.com | LinkedIn | GitHub

Summary

Full Stack Engineer with **2.5+ years of experience** delivering production web applications across **React/Next.js** frontends and **Node.js/FastAPI** microservices. Experienced in building high-throughput REST APIs, real-time WebSocket pipelines, and integrating applied AI retrieval systems (Qdrant, pgvector). Proven record of reducing API latency by **60%** via database query indexing, establishing automated test coverage, and deploying reliable cloud services on **AWS**.

Technical Skills

Frontend Engineering: React.js, Next.js, TypeScript, JavaScript (ES6+), Redux Toolkit, Tailwind CSS, Vite, WebSockets

Backend & APIs: Node.js, Express.js, NestJS, Python, FastAPI, AsyncIO, REST APIs, Microservices, Idempotent Webhooks

Databases & Caching: PostgreSQL, Redis (Caching, Task Queues), MongoDB, pgvector, Compound Indexing, Connection Pooling

Cloud, DevOps & Quality: AWS (EC2, S3), Docker, GitHub Actions CI/CD, Jest, Postman API Testing, Linux, Nginx

Applied AI & Retrieval: RAG Pipelines, Vector Search (Qdrant, pgvector), Hybrid Search (BM25), Semantic Caching, Gemini & OpenAI APIs

Experience

AI Developer – SahioNext Technology Pvt. Ltd. — Oct 2025 – Apr 2026

Remote

- Architected an end-to-end full-stack media platform using **FastAPI, React.js, and PostgreSQL**, handling asynchronous job dispatching for **10,000+ audio sessions** via **Redis queues and background workers**.
- Engineered real-time operational telemetry dashboards in **React.js and Redux Toolkit** to monitor token expenditure, rate-limit thresholds, and pipeline bottlenecks, cutting debugging triage time from hours to minutes.
- Built full-duplex **WebSocket ingestion pipelines** with exponential backoff and automatic session recovery, maintaining stable real-time audio evaluation streams over high-latency client networks.
- Integrated semantic search microservices using **Qdrant and FastAPI** (¡300ms p95 latency) over **10,000+ indexed chunks**, surfacing relevant lesson transcripts for automated CEFR language assessment.

Full Stack Developer (Contract) – PearlThoughts — Feb 2025 – Sep 2025

Remote

- Engineered multi-tenant backend automation microservices using **NestJS, TypeScript, and Google Workspace APIs**, orchestrating **1,000+ daily business workflows** and eliminating manual client data entry.
- Built resilient event-driven pipelines with **exponential backoff, idempotent webhook handlers, and rate limiting**, ensuring zero duplicate transactions across concurrent tenant processes.
- Developed automated test harnesses and isolated **Docker autograder environments** for enterprise **Coursera course launches**, preventing runtime execution regressions prior to production release.

Full Stack Developer – The DevOps Team — Oct 2023 – Jan 2025

Bangalore, India

- Optimized high-traffic **Node.js and Express.js APIs** using **Redis caching, PostgreSQL compound indexing, and connection pooling**, reducing median response times from **800ms to 320ms (60% decrease)**.
- Implemented automated integration and unit test suites using **Jest and Postman**, ensuring regression-free deployments across core authentication and search endpoints.
- Built responsive onboarding and profile workflows in **React.js and Next.js**, running A/B experiments on multi-step forms to lift user registration conversion by **18%**.

Projects

WorkBridge – Full-Stack Collaboration Platform — *May 2026 – Present*

Next.js, React.js, TypeScript, Node.js, Express.js, FastAPI, PostgreSQL, pgvector, Redis, Tailwind CSS | Live Project

- Designed and implemented a decoupled microservices platform featuring a **Node.js/Express API gateway** for session/auth handling and an asynchronous **Python/FastAPI microservice** for vector processing.
- Developed dynamic, responsive client-side interfaces using **Next.js, React.js, and Tailwind CSS**, featuring real-time project collaboration boards, role-matching filters, and optimistic UI updates.
- Enforced role-based access control (RBAC) and OAuth2 authentication backed by **PostgreSQL and Redis**, achieving **¡150ms p95 latency under k6 benchmark load tests**.

Certifications

Oracle Agentic AI Certified Foundations Associate – Oracle — Aug 2026
Oracle AI Vector Search Certified Professional – Oracle — Sep 2025
Machine Learning I – Columbia University — Aug 2025
Oracle Cloud Infrastructure Generative AI Professional – Oracle — Aug 2024

Education

Adichunchanagiri Institute of Technology (VTU) — Chikkamagaluru, KA, India
Bachelor of Engineering — *Aug 2023*